

Solution Requirements

Date	1 June 2026
Team ID	XXXXXX
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>[Describe how users will securely log in or verify their identity]</i>	Medium
2	Authorization levels	<i>[Define the user roles and what access/permissions each role has]</i>	Low
3	External interfaces	<i>[Detail any third-party APIs, external databases, or]</i>	High

		<i>services the system must connect to]</i>	
4	Transactions processing	<i>[Explain how data exchanges, payments, or core system transactions will be handled]</i>	High
5	Reporting	<i>[Describe the types of reports, dashboards, or data exports the system must generate]</i>	Medium
6	Business rules, etc.	<i>[List specific logic, constraints, or workflows unique to the business process]</i>	High
7	Compliance to laws or regulations	<i>[Specify any legal, industry, or data privacy standards the system must adhere to (e.g., GDPR, HIPAA)]</i>	Medium
8	Other	<i>[Add any additional functional requirements here]</i>	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	<i>The system should provide crop recommendations quickly.</i>	<i>Prediction generated in less than 2 seconds</i>
2	Scalability	<i>The application should support multiple users simultaneously.</i>	<i>Supports 100+ concurrent users</i>
3	Security & Data Privacy	<i>User input data should be processed securely without unauthorized access.</i>	<i>Secure data handling and protected access</i>
4	Reliability & Availability	<i>The application should remain available with minimal downtime.</i>	<i>99% uptime</i>
5	Usability & Accessibility	<i>The interface should be simple, responsive, and easy to use.</i>	<i>User-friendly design compatible with desktop and mobile devices</i>
6	Other	<i>The system should be maintainable and allow future enhancements.</i>	<i>Modular code for easy maintenance and updates</i>